

AUCell-style gene-set scores in sparse single-cell data: a closed-form confound, matched nulls, and where they stop working

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Abstract

AUCell-style gene-set scores ask whether a gene falls in a cell's top-ranked block. In sparse data a cell detects fewer genes than the block holds, so the surplus ranks are settled by a tie-break among genes it never expressed. That component is closed-form in the depth vector alone, so any axis tracking depth acquires an association with the score owing nothing to biology. Under a true null over 1,080 runs, cell-level correlation, label permutation, median split and outcome-chosen cutpoints rejected at 0.400, 0.397, 0.337 and 0.556; the first rose to 0.694 under confounding while a detection-matched null held near nominal. A product of two correlations, both computable before any outcome is seen, forecasts the inflation across its range and recovers the realised association at 0.940. The matched null holds only while it reproduces the set's co-detection structure; we measure where it does not.

Keywords: gene-set scoring; single-cell RNA-seq; rank statistics; sequencing depth; matched null; reproducibility

1 Introduction

A gene-set score reduces a cell's transcriptome to one number. The family that does so by ranking — AUCell [1], UCell [2], GSVA [3], singscore [4], the module score of Tirosh et al. [5], and the enrichment scores of Barbie et al. [6] and Subramanian et al. [7], several collected behind one interface by decoupleR [8] — asks where the set's genes fall in the cell's ranked expression, and summarises that with an area or a rank statistic. Scores in this family are now a routine readout in single-cell studies, from targeted signature tests to atlas-scale profiling [9]: they are used to call cell states, to order cells along axes, and increasingly to carry a statistical claim, since the score is continuous and the axis of interest is continuous and a correlation between them has a p-value attached. The methods have been compared with one another on accuracy, stability and cost [10], in the broader tradition of scoring single-cell methods against known ground truth [11]; what has not been characterised is the null that the p-value is computed against.

The arithmetic below covers the scores that summarise a fixed top block of the ranking: AUCell, the canonical case, and UCell, which fills ranks the same way. GSVA, singscore and the module score truncate nothing and are not measured here (Supplementary Table S6).

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That last step is where the difficulty lies. The hazards usually named for single-cell statistics — pseudoreplication across cells from one donor [12, 13], zero inflation [14, 15], the unreliability of pathway scores on shallow data [16] — are hazards of the design or of the data. This one is a property of the score itself, deterministic given the depth vector, computable before the gene set is scored or the outcome examined.

The mechanism is a tie-break. A rank-based score places each gene at its rank within the cell. If the cell detects D genes and the score's rank ceiling is m , and $D < m$, then the positions from $D+1$ to m have no counts to order them: every gene with a count of zero is tied, and the implementation breaks the tie arbitrarily. A gene the cell never expressed therefore enters the top block with probability $(m - D)/(G - D)$, where G is the panel size; the implementation breaks the tie arbitrarily, an arbitrariness that measurably moves the score when the tied order is permuted [17]. In the breast-cancer dataset we use throughout [18], the panel has 24,646 genes and the AUCCell default ceiling is 1,233; CD8⁺ T cells detect a median of 882. For those cells the ceiling sits well above everything the cell expressed, and most of the "ranking" is a coin toss.

Because $(m - D)/(G - D)$ falls as D rises, the tie-break's contribution to the score is a decreasing function of depth. That alone would be harmless if depth were noise. It is not: depth tracks almost every quantity a single-cell study measures, from cell-cycle phase to exhaustion, because RNA content is biology — and because the cells it starves are the ones quality control flags [19] — and it is entangled with biology tightly enough that correcting it out degrades rather than improves a spatial analysis [20]. An axis that correlates with depth therefore correlates with the score through the tie-break, and the conventional tests — a correlation across cells, a permutation of the axis across cells, a split at the median — are all evaluated against a null in which cells keep their own tie-break contributions. The confound is not shuffled away, because it is inside every cell.

This paper does three things. It gives the closed form, so that the size of the effect is a calculable property of the matrix rather than a surprise. It measures the operating characteristics of seven analyses — four conventional, three with matched nulls — on a ground-truth grid in which the truth is known by construction. And it reports, with the same precision, where the matched nulls stop working: a framework that claims universal validity is less useful than one that states its boundary.

2 Results

2.1 The channel, and how open it is

Let the panel contain G genes, let cell c detect D_c of them, and let $m = \lceil fG \rceil$ be the rank ceiling ($f = 0.05$ for AUCCell). Write $d_c(S)$ for the number of genes of set S that cell c detects and $k = |S|$. Two quantities follow from the arithmetic of tie-breaking.

A gene the cell does not express lands inside the top m with probability

$$\pi(D_c) = \frac{m - D_c}{G - D_c}, \quad (1)$$

and its expected share of an AUCCell score is

$$\tau(D_c) = \frac{(m - D_c)(m - D_c + 1)}{2m(G - D_c)}. \quad (2)$$

The score then splits into a part the cell's expression supports and a part the tie-break supplies,

$$\text{AUC}_c = \text{Det}_c + \frac{k - d_c(S)}{k} \tau(D_c) + \varepsilon_c, \quad (3)$$

an exact decomposition into signal, confound and residual. The exact part is the split itself: the score is, cell by cell, the sum of what the detected genes contribute and what the tie-break realises, and the package’s test suite asserts that equality cell by cell. The closed form $\tau(D_c)$ is the expectation of the tie-break part under the uniform-order model (2), so ε_c is the realised tie-break centred on its expectation — and because the hash that realises the tie-break is deterministic, ε_c is a reproducible property of the matrix rather than sampling noise. Measured over the calibration grid’s own configurations (90 datasets, 108,000 cells), it is centred where the model says it should be (mean +0.0002), of size 0.012 in standard deviation (0.019 at the sparsest configuration), and it carries no depth signal of its own: its correlation with D_c stays between -0.06 and +0.04 across every configuration. What depth contributes to the score is the middle term, and Fig. 1a draws its two ingredients on one axis: the closed-form probability (1) across the depths a real experiment contains, the ceiling it is evaluated against, and every compartment of the benchmark at its own median depth.

The middle term of (3) is the confound. It depends on the cell only through D_c and on the gene set only through $k - d_c(S)$, so it is shared by every gene set with the same detection profile — and it is larger, in absolute value, in cells that detect less. How much of a set’s score is supplied by this term rather than by expression is its *tie-break share*. With $\bar{\pi}$ the mean of (1) over cells and $\bar{p} = \overline{d_c(S)}/k$ the set’s mean detection rate, it is

$$\text{share}(S) = \frac{(1 - \bar{p}) \bar{\pi}}{\bar{p} + (1 - \bar{p}) \bar{\pi}}. \quad (4)$$

Equation (4) is a property of the set in the matrix. It needs no outcome, no score and no null, and it is the first thing the framework reports. It describes the mechanism by which sparsity enters a score. Whether it also predicts the magnitude of the resulting artefact is a separate question, and not an obvious one — a large tie-break share can mean a large artefact or a score whose variance is so dominated by tie-breaking that the artefact is attenuated by it. Section 2.3 answers that question by measurement rather than by assumption.

What the magnitude follows from is the score’s own dependence on depth. If the score tracks depth with correlation $\rho_s = \rho(\text{AUC}, D)$ and the axis under test tracks depth with $\rho_a = \rho(\text{axis}, D)$, the two induce an association between score and axis that has nothing to do with biology, of magnitude

$$\rho_{\text{implied}} = \rho_s \rho_a. \quad (5)$$

This is the classical attenuation-of-confound identity, and it is what (3) produces when the tie-break term is summed over cells. Both factors are computable before any outcome is examined: ρ_s from the matrix and the gene set, ρ_a from the axis the analyst intends to test. The package’s `preflight()` estimates the same quantity structurally, from the detection probabilities alone, via the τ of (2); the two routes are compared in Section 2.3.

2.2 Conventional tests reject in proportion to the channel

We built a simulator with an explicit confounder, so that the truth is known. General-purpose single-cell simulators [21, 22] generate counts and leave the scoring to the user; here the scoring step is the mechanism under test, so the simulator couples the two. It generates a depth axis that depends on a latent programme, a target gene set whose expression depends on depth, and a background of co-expressed modules. Under `effect = 0` the set has no relationship with the programme whatsoever: every rejection is a false positive with respect to the claim the analysis is making.

Grid A varies the confounding (programme-to-depth loading 0, 0.3 and 0.6), the set’s sparsity, the background co-expression and the cell depth, for 1,080 runs. Table 1 gives the false-positive rate of each analysis at $\alpha = 0.05$.

Table 1: False-positive rate under a true null, by analysis. Grid A, 1,080 runs, `effect` = 0. Intervals are Wilson score intervals [23]. A valid test sits at 0.05. The four conventional analyses are anti-conservative; the three matched families are closer to nominal and, as Table 2, do not worsen with confounding.

Analysis	Null	Rate	95% CI
Cell-level correlation	conventional	0.400	[0.371, 0.430]
Label permutation	conventional	0.397	[0.368, 0.427]
Median split	conventional	0.337	[0.309, 0.366]
Cutpoint chosen from outcome	conventional	0.556	[0.527, 0.586]
Size-matched null	matched	0.201	[0.178, 0.226]
Expression-matched null	matched	0.068	[0.054, 0.084]
Co-detection-matched null	matched	0.094	[0.078, 0.113]

Table 2: The conventional failure is the confounder, and the matched failure is not. Rows are the planted programme-to-depth loading in grid A; each level contributes 360 runs. The cell-level rate rises from 0.064 to 0.694 with the confounder. The expression-matched rate does not rise: 0.069, 0.072, 0.061.

Programme–depth loading	Cell-level	Outcome cutpoint	Expression-matched	Co-detection-matched
0 (controls)	0.064	0.308	0.069	0.075
0.3 (intermediate)	0.442	0.586	0.072	0.083
0.6 (strongest)	0.694	0.775	0.061	0.125

The four conventional analyses reject an association that does not exist at 40.0%, 39.7%, 33.7% and 55.6% of runs. The failure is not uniform: it is absent where there is no confounder and severe where there is one. Table 2 gives the same rates by the strength of the planted programme-to-depth coupling.

With no confounder planted, the cell-level test is at 0.064, near nominal. With the strongest, it is at 0.694. Label permutation does not help — it is within a percentage point of the cell-level test at every level — and the reason is visible in (3): permuting the axis across cells destroys the axis’s correlation with depth in the permuted draws, while the observed statistic keeps it. The null is centred on zero and the observation is not, so the test rejects.

The matched families behave differently. The expression-matched rate is 0.069, 0.072 and 0.061 at the three levels: it does not respond to the confounder at all. That is the central result, and it is the right shape. Validity under a confounder means a rate that does not move when the confounder moves.

2.3 The diagnostic, and a diagnostic that failed

A rate that climbs with a latent variable is not yet a usable warning. The framework claims something stronger: that a quantity computed from the matrix and the gene set, before any outcome is examined, tells the analyst where on that climb they are.

We first proposed (4), the tie-break share, for that role. It does not work, and the failure is instructive. Binning the 1,080 runs of grid A by the share, the naive rejection rate runs 0.583 at the lowest share to 0.283 at the highest (bins 0.003 to 0.703). The curve is not flat — which is what a merely uninformative quantity would give — and it is not monotone: it goes the *other* way, the highest-share sets being the ones the test handles best. A Cochran–Armitage test for trend gives $z = -8.73$ ($P = 2.6 \times 10^{-18}$), significant and in the direction opposite to the one the framework predicted; the permutation and optimal-cutpoint tests reverse with it.

The reason is a distinction between mechanism and magnitude, and it is visible in (3). The share

measures how much of a set’s score the tie-break supplies. That is the mechanism by which depth enters the score, and it is why the confound exists at all. It is not the magnitude of the artefact. A set sparse enough to have a large share is one whose score is mostly tie-breaking, and a score that is mostly tie-breaking is mostly the tie-break’s own cell-to-cell fluctuation, the ε_c of (3); summed over cells that fluctuation attenuates the very association the share was meant to forecast. Among the sparsest sets, then, a higher share buys more mechanism and less signal at the same time, and past a point the second effect wins. The share is a correct description of the channel. It is not a forecast of the damage.

The magnitude is forecast by (5), which asks the two questions that determine it: does the score track depth, and does the tested axis track depth. Across the 1,080 grid-A runs the product recovers the realised association almost exactly — Pearson 0.940, slope 1.057, mean predicted $+0.0326$ against mean realised $+0.0344$. Binned by its magnitude, the same runs give a curve that behaves: 0.078 in the lowest bin, rising monotonically to 1.000 at the top ($z = +23.71$, $P = 2.6 \times 10^{-124}$), with the permutation and optimal-cutpoint tests following it to 0.994 and 0.994. A quantity the analyst can compute before looking at a single outcome separates runs where the test is perfectly calibrated from runs where it is certain to reject.

The matched nulls are flat on the same axis, which is the stronger of the two results: where the conventional tests climb to certainty, the expression-matched family stays at 0.089 in the lowest bin and 0.083 in the highest ($z = +0.32$, $P = 7.5 \times 10^{-1}$, not significant), and the random-matched family 0.072 to 0.717 ($z = +15.85$, $P = 1.4 \times 10^{-56}$). The fix works exactly where the diagnostic says the problem is, and the diagnostic says there is no problem where the fix is not needed.

The sharpest evidence that (5) is the right quantity and (4) the wrong one comes from the null whose construction guarantees no confound. On the random-matched family, the share reports a highly significant trend in the rejection rate ($z = -11.58$, $P = 5.0 \times 10^{-31}$) where the product reports none. That is a diagnostic raising an alarm about a confound that cannot exist — the failure mode that would make an analyst doubt a sound result — while missing the confound that does.

One matched family retains a real trend on this axis: the co-detection-matched null rises from 0.094 to 0.183 across the bins, $z = +3.29$. It is the family that fails at the boundary described next, and its residual is the boundary seen from the other side.

Binning is a coarse comparison, so both candidates were also put in the same model over all 1,680 effect-free runs of the two sweeps. The share’s coefficient is indistinguishable from zero once the product is present, and on its own it is -0.184 ($P = 4.2 \times 10^{-4}$) — not merely uninformative but negative. The share is not a weak version of the right quantity; it is a different quantity. The fit is reported in the Methods.

2.4 The curve replicates on an independent sweep

Grid A varies the confounder by construction. Grid B varies something the analyst can see — the ratio of the rank ceiling to the compartment’s median detected count, swept from 0.43 to 6.38 — and it runs on a different set of 600 datasets. Where the ratio is below one the ceiling sits inside what the cells expressed and there is no tie-break to be confounded by; where it is well above one the top block is mostly zeros. Splitting the sweep there gives the cleanest statement of the mechanism in the paper: the cell-level test rejects at 0.182 with the channel shut and 0.442 with it open, over 170 and 430 datasets, while the matched null does not move (0.059 against 0.070).

The response curve against (5) reproduces on this sweep rather than restating the grid-A result: the cell-level rate trends at

$$z = +19.06$$

Table 3: The residual belongs to the set under test, not to the null. Under `effect = 0` with a strong confounder, the expression-matched test is applied to the simulator’s own target sets and to sets drawn from the null’s generator. The exchangeable sets sit at nominal; the targets with planted co-expression do not. Each value is a false-positive rate at $\alpha = 0.05$ shown with the number of sets it was computed over; the interval for the rate nearest the nominal level is given in the text.

Configuration	Target sets	n	Exchangeable draws	n
Co-expressed, confounded	0.125	24	0.026	192
Sparse, confounded	0.042	24	0.078	192
Unconfounded control	0.000	24	0.026	192

, the outcome-chosen cutpoint at

$$z = +13.95$$

, and the two matched families at

$$z = +3.07$$

and

$$z = +1.17$$

, neither significant. A trend that survived only on the grid it was discovered on would not be worth stating; this one transfers to different factor levels, a different range and different datasets. Two further results from the sweep — the floor the outcome-chosen cutpoint does not fall below, and the instability of the tie-break share’s apparent trend — are reported in the Methods. Fig. 1c draws both sweeps together: above, the rejection rate binned by (5); below, the same runs binned instead by the share, which falls where the argument required it to rise.

2.5 Where the matched null stops working

We did not expect to find a boundary, and reporting it is the reason this paper is worth more than its headline. Two failures survive in the matched families, and they are of different kinds. Fig. 1d collects all seven analyses side by side, split by whether there is a channel to be confounded by.

The first is small and constant. The expression-matched family sits at 0.068 rather than 0.05, and the excess does not grow with the confounder (Table 2). The grid’s co-expression factor separates it: at `coexpr = 0` the expression-matched rate is 0.056 over 540 runs, a Wilson interval that contains the nominal level (yes); at `coexpr = 0.6` it is 0.080, and the interval excludes it (no). So the residual is not a constant offset of the method; it appears exactly where the gene set carries internal structure.

To establish that this structure, and not the matching, is responsible, we ran an audit that adds a third kind of set to the same test: *pseudo-targets*, drawn from the null’s own generator and exchangeable with the draws by construction. A test that is anti-conservative for the simulator’s targets and nominal for pseudo-targets is indicting the targets, not itself.

The exchangeable sets sit at nominal everywhere: 0.026 where co-expression is planted, 0.078 where it is not — narrowly, the interval running 0.048 to 0.125, which is what the numbers say and all that should be made of it — and 0.026 in the unconfounded control. The simulator’s targets do not: 0.125, 0.042 and 0.000 respectively. Where the sets carry structure the draws cannot reproduce, the target is rejected 4.8 times as often as a set exchangeable with the draws; where no such structure is present the two are indistinguishable at these counts. The residual the framework reports is therefore the co-expression channel, and against the draws it was built to replace the matched null is doing what it claims.

Table 4: The co-detection-matched null fails where it failed to match. Grid A, effect = 0. “Gap” is the shortfall of the draws’ co-detection below the query’s, as defined in the text. Neither column uses the outcome.

Rule	n flagged	Flagged rate	n not flagged	Unflagged rate
gap > 0.005, $ \rho > 0$	385	0.153	695	0.062
gap > 0.005, $ \rho > 0.2$	262	0.187	818	0.065
gap > 0.010, $ \rho > 0.2$	221	0.204	859	0.066
gap > 0.020, $ \rho > 0.2$	136	0.228	944	0.075

That is a useful thing to know but not yet an actionable one, because the analyst does not know in advance whether their set is such a set. The second failure is the actionable one. The co-detection-matched family is the most elaborate of the three: it draws candidates matched on per-gene detection and selects the one whose mean pairwise co-detection is closest to the query’s. On the grid as a whole it rejects at 0.094, worse than the simpler expression-matched family. But splitting by the mismatch it actually achieved separates it cleanly from its own successes, and both quantities that do the separating are computable before any outcome is examined — the achieved co-detection gap, which the constructor returns, and the correlation between the tested axis and the cell’s detected-gene count, which is a property of the data.

Where the gap is small the co-detection-matched null is at 0.062 — nominal. Where the gap exceeds 0.005 and the axis tracks depth it is at 0.187, and at 0.228 once the gap exceeds 0.020. The two conditions multiply, which is what the mechanism predicts: a co-detection mismatch makes the drawn sets less internally correlated than the query, so their scores are less dispersed than the query’s, so the query’s correlation lands in their tail more often than it should; and the confounder is what gives the query’s correlation a magnitude worth landing on. Neither condition alone is enough, and neither requires the outcome.

This is the framework’s boundary, stated as a rule. The matched null is valid when the confound the product (5) measures is small *and* the null reproduces the set’s co-detection structure. When the second condition fails, no family in the framework is safe, and the honest output is a refusal rather than a p-value.

2.6 Dichotomisation costs more than the confound does

Splitting a continuous score at a cutpoint is a separate decision from how the score is nulled, and the two compose. A median split is mildly anti-conservative on its own (0.337); a cutpoint chosen to maximise the association — the **optimum** rule that survival and response analyses use routinely — is worse than any other analysis in the table at 0.556, and at 0.775 under the strongest confounder. The penalty is not a small additive constant: at every level of confounding the outcome-chosen cutpoint costs 16.4 to 26.4 percentage points over the median split, and the prognostic-factor literature on this hazard [24, 25] applies unchanged.

We make the measurement available as `cutpoint_fpr`, which permutes the outcome and re-selects the cutpoint in every permutation, so that the reported rate is the rate of the analyst’s whole procedure rather than of the test statistic alone.

2.7 Donor structure is a second channel with a different remedy

The hazard above is a property of the score. This one is a property of the design, and the two are worth separating because they are routinely conflated and their remedies are not the same. Grid E crosses the number of donors with the confounder and plants no biological effect, holding the

sparsity and co-expression factors where the tie-break channel is not the explanation, so that what the cell-level test reports there can be attributed to the donor structure.

With 1, 4 and 16 donors the cell-level test rejects at 0.300, 0.375 and 0.237.

The remedy is not a null. Every family in this framework draws cells, and a null that draws cells inherits the dependence it was meant to remove: the expression-matched test gives 0.037, 0.037 and 0.025 on the same runs, tracking the test it was built to replace rather than the nominal level. Resampling within a compartment cannot manufacture donors, and neither can any permutation of the axis across cells. The resampling unit has to become the donor, which changes the analysis rather than the null, and that literature is separate and already established [12, 13]. The separation has a consequence for reading this paper: the calibration grid is run with donor structure switched off, so the inflation it measures belongs to the tie-break and cannot be pseudo- replication in disguise.

2.8 What validity costs

A null that never rejects is trivially well calibrated, so the price of calibration has to be stated for the calibration claim to mean anything. Grid C plants an effect — 0.25 and 0.5 standard deviations on the axis — over the same sparsity and confounder factors, and asks each analysis to detect it. At the smaller effect the cell-level test detects 0.829, the expression-matched null 0.871 and the co-detection-matched null 0.863; at the larger effect the three give 0.946, 0.988 and 0.988.

The comparison that matters is not against the cell-level test, which is inflated, but against what the matched nulls give up. Where the channel is shut the two nulls are close to the test they replace, and where it is open the cell-level test’s own rate is not a detection rate — the runs it "detects" in are the runs whose null was wrong — so the honest comparison at those settings is between the two matched families.

2.9 Real data

Every measurement so far is on matrices the simulator built, so the remaining question is whether the regime it describes is one real analyses occupy. We scored 1,374 gene sets from 6 MSigDB collections (h.all, c2.cp.kegg legacy, c2.cp.biocarta, c2.cp.reactome, c2.cp.wikipathways, c7.immunsigdb) [26] across the 16 annotated cell types of GSE176078 [18], a 55,003-cell, 24,646-gene breast-cancer atlas, computing for every (compartment, set) pair the ceiling-to-depth ratio, the tie-break inclusion probability, the realised share of the score that came from ranks the tie-break settled, and the correlation of the score with the cell’s detected-gene count. Fig. 1b places the first two of those against each other, one point per compartment.

The channel is open across most of the cohort. A median of 0.04 of each score came from ranks the tie-break settled rather than from counts the cell observed, the lowest compartment median is 0.01 and the highest 0.12. In CD8⁺ T cells — the compartment the introduction quotes — the median cell detects 882 genes against a ceiling of 1,233; 7 of the 16 compartments have the channel open on the paper’s criterion and 9 do not.

The regime quantity predicts what the regime does, at both levels the benchmark resolves. Across compartments, the tie-break inclusion probability correlates with the compartment’s median absolute score-depth correlation at +0.71. Those medians run from 0.14 to 0.27, so the 0.3 line that separates alarming from tolerable at the level of a single (compartment, set) pair has no compartment on its far side: the inclusion probability orders compartments, it does not partition them, and we report the rank correlation rather than an AUC that would have one class and no contrast to score. Within a compartment, the realised share of the score correlates with the absolute score-depth correlation at -0.63 — whereas the detector proposed in an earlier version of this work, the median per-cell detection rate, gives +0.32. Per pair rather than per compartment the line does have both sides:

over the benchmark as a whole 32.0 of (compartment, set) pairs reach $|\rho| \geq 0.3$ against depth and 8.0 reach 0.5.

The channel is not a property of one family of gene sets, and the collection structure of the benchmark is what shows that rather than assumes it. Grouping each cohort’s sets by their six source collections and asking, inside every compartment separately, how a set’s median detection rate orders its score–depth association, the diagnostic keeps its sign in all twelve collection-by-cohort groups, running from +0.26 to +0.73; no family inverts the ordering the pooled benchmark reports. What varies between families is the level of the damage: collection medians of $|\rho|$ run from 0.16 to 0.50 across the two cohorts, and the named sets a reader would check first span that range. The large co-regulated modules move up with the second cohort’s greater exposure — `HALLMARK_HYPOXIA` from 0.47 to 0.65, `KEGG_CELL_CYCLE` from 0.30 to 0.56, `HALLMARK_INTERFERON_GAMMA_RESPONSE` from 0.49 to 0.53— while the two small checkpoint sets sit below the 0.3 line in both cohorts (PD-1 from 0.26 to 0.20, CTLA-4 from 0.22 to 0.18). Supplementary Tables S2 and S3 give the cohort characteristics and the per-collection and per-set view.

The same arithmetic governs the other implementation. UCell’s documentation recommends setting its rank ceiling near the median detected genes per cell and ships a default of 1500 for 10x data; where the compartment’s median detection lies outside that default’s reach the two disagree, and in 23.9 of pairs the score computed under the default is a materially different number from the score computed under the package’s own recommendation. The disagreement is not random: the tie-break inclusion probability, which needs only the depth vector, flags it.

The same measurements repeat in an independent cohort, and they divide cleanly between what the channel does and what the biology does. Applied to GSE161529 [27] — 17 clusters, 29,927 cells scored at a median of 1,258 per cluster, the 33,538 genes measured — the pipeline sets a ceiling of 1,677 rather than the first cohort’s 1,233, because the ceiling is a fraction of the panel and the same gene set therefore lands on a different score scale in each. Two scores are not comparable as values, and what is compared is direction. The regime itself reproduces, and this is the more open of the two cohorts: a median detection of 1073 genes against a median tie-break inclusion of 0.017, and a median score–depth correlation of +0.31.

The association is the part that does not travel. Across the 1,077 gene sets scored in both cohorts, the per-set association with the compartment’s dominant axis agrees between the cohorts at +0.10, with 57% of sets keeping its sign and a median move of 0.06 — figures dominated by sets whose association is near zero in both, where the sign is decided by noise. Among the 70 sets whose association exceeds 0.20 in the first cohort, 83% keep its sign in the second, while the association itself falls from a median of 0.25 to 0.04. Direction survives a change of panel, annotation and cell number; magnitude does not, which is the sense in which the checklist’s external tier is a test of direction rather than of effect size.

3 Discussion

The tie-break in AUCell-style gene-set scores is not a subtle statistical pathology that needs care to detect. It is arithmetic: a cell that detects fewer genes than the ceiling has empty ranks, and something has to fill them. What has been missing is not the observation but the calibration — a statement of how much the resulting component contributes, under what conditions it turns into a false positive, and where matching on composition stops being enough. The nearest prior work measures the variability and recommends aggregating cells [17]; the closed form explains both: pooling raises the detected count above the ceiling and the fill collapses. The tie-break is a property of the block summary, not of ranking itself: enrichment methods that weight the whole ranked list [7, 28] have no ceiling to fill.

Our answer to the first two is quantitative and, we think, decisive. Under a true null the conventional analyses reject at 40.0% and rise to 0.694 as the confounder strengthens; a null matched on per-gene detection does not move. The pre-flight quantity (5) needs no outcome — only the depth vector, the score and the axis — so the practical recommendation is simple: compute the product before the test, and where it is not small, do not report a conventional p-value at all. The quantity that first suggested itself for that role, the tie-break share, is refuted above, and the reason it fails is the reason the product succeeds: the share measures the channel, and what has to be forecast is the damage.

Our answer to the third is the part we would most want a reader to take away. Matching on composition is not a universal fix. It fails when the gene set carries internal correlation that the null does not reproduce, and the failure is large — up to 0.228 in the grid — and, crucially, predicted by quantities that are measured anyway. A methods paper that reports its own boundary is more useful than one that does not, and the boundary is not a technicality: real gene sets, especially co-regulated modules, have exactly the internal structure the failure requires.

The rule has two edges, and the cheap one is easy to miss. Where the pre-flight product sits inside the chance band the conventional null is valid, and the framework’s only output is the certificate that nothing needs doing — no matched null, no re-analysis. That gate earns its cost when the alternative is to run the expensive machinery everywhere or to trust it nowhere. What the framework does not do is remove the confounder. Matching stops a false positive; it does not turn a depth proxy into biology. Nor does the ordinary correction touch the channel: size-factor normalisation [29–32] rescales counts but changes no count from zero to non-zero, so the tie-break’s dependence on the detected-gene count survives it. Where the question is itself depth-dependent, the remedy is design — an axis that does not track depth — and inference cannot supply it. Supplementary Table S5 states the decision points as the software applies them, thresholds included.

Four limitations bound what we have shown. The grid is a simulation, and while its parameters are set from the real compartment it is calibrated on, simulation always understates the variety of real data. The co-expression structure we vary is a single planted module; real co-expression is hierarchically organised, and the co-detection gap is a scalar summary of it, so the boundary we measure is sharper than the boundary a user will meet. The framework’s verdict rule thresholds are choices, reported as such, rather than quantities derived from first principles. And what an external replication can carry is direction: a second cohort reproduces the channel, whose arithmetic is a property of the panel and the depth distribution, but not the association — of the 70 sets clearing $|\rho| = 0.20$ in the first cohort, 83% keep the sign in the second, and the association itself falls from a median of 0.25 to 0.04. The checklist’s external tier is written for exactly this: it compares direction and rank association rather than values, and a set that replicates in direction should be expected to do so at a different magnitude.

Two extensions follow directly. The first is a null that reproduces the set’s internal correlation structure rather than approximating it with a scalar: matching on the set’s leading covariance directions, or drawing by rotating labels within co-expression modules, would remove the residual we characterise in Table 4. The second is to make the boundary automatic — for the checklist to refuse a matched p-value whenever the gap exceeds tolerance and the axis is depth-linked, which changes a threshold rather than the method and is the next thing we will do.

4 Methods

4.1 Scoring

Scores are AUCell [1] computed as the area under the recovery curve, with the ceiling $m = \lceil fG \rceil$ and $f = 0.05$ unless stated — the ceiling the released package also ships as its default. Where the other ranking tools sit relative to this arithmetic is tabulated, with the source of each statement marked, in Supplementary Table S6. The rank cache stores, for each cell, the ranks of the genes retained for scoring, clipped at a ceiling; ranks are computed within a cell and are therefore invariant to which other cells are present, which is what makes a cache over one compartment give the same answer as a cache over all cells. Ties among undetected genes are broken by a deterministic hash of the cell and gene indices, so that the same matrix always yields the same score and two implementations can be compared to the integer. The residual ε_c of equation (3) is measured, not assumed: across the calibration grid’s own configurations it reaches its 99.9th percentile at 0.13 and its largest observed value is 0.16, in the sparsest configuration, while the realised split reproduces the score to 1.1×10^{-16} .

4.2 Matched nulls

Three families are drawn, each excluding every gene of the query set. **random** matches set size only and is the naive baseline. **expression** matches per-gene detection rate within strata, refined by relative distance in mean expression. **codetection** draws expression-matched candidates and selects the one whose mean pairwise co-detection, computed as the cosine similarity of the detection vectors, is closest to the query’s. Every family reports the composition it achieved — detection ratio, expression ratio, co-detection gap and the overlap with the query — and the detection floor is derived from the query’s own detection rather than fixed, because a fixed floor censors the candidate pool from below and makes every replacement denser than the gene it replaces.

4.3 Tests

The cell-level test is Spearman’s rank correlation [33] between score and axis across cells. The permutation test shuffles the axis across cells. The median and outcome rules dichotomise the score at the median and at the cutpoint maximising the association. Matched p-values are Monte-Carlo with the $(k + 1)/(n + 1)$ correction, two-sided about the centre of the null’s correlation, which is the question a user of the framework asks: whether the set is associated with the axis, not whether it is depleted by it. The two directional values are reported beside every matched p-value, and every matched rejection rate in Supplementary Table S1 is recomputed in each of the three tails, so that no conclusion here rests on the choice of tail. It matters that the convention is stated: an earlier version of these grids tested the lower tail alone, which halved every matched rejection rate and reported the matched null’s power against a planted effect as zero.

4.4 Simulation

The simulator generates a depth factor per cell that depends on a latent programme, a target set of k genes whose detection rate follows a specified value, and a background of co-expressed modules. Parameters are the panel size, the median detected genes per cell, the set’s detection rate, the programme-to-depth loading, the effect of the programme on the target genes, the co-expression loading, and the number of donors. One run is one record: a single JSON line in `results/grid_*.jsonl` carrying the design factors, the seed, the realised depth statistics and every test outcome, written as the run finishes rather than collected afterwards. The five grids hold 2,800 such records in all,

and every rate and table in this paper is generated from those records by `analyse_grid.py` and `make_numbers.py`, so a table and its records cannot disagree.

4.5 Grids

Grid A is a full factorial of confounding, sparsity and co-expression — 18 configurations at 60 replicate datasets each, 1,080 runs, all with `effect = 0`. Grids B–E vary the ceiling-to-depth regime, the planted effect, the set size and the number of donors respectively. Each grid holds donor structure out except where it is the factor under study, so that what the other grids measure is the tie-break channel rather than the donor channel. Two columns that look like design factors are not: the achieved median detected count and the resulting ceiling-to-depth ratio are produced by the simulation, so they are reported and stratified on but never used to define a cell of the design. Supplementary Table S7 states each grid’s factor levels, held-fixed factors, configuration counts and seed range, computed from the records themselves.

Rejection rates use a nominal $\alpha = 0.05$ and are reported with Wilson score intervals. Those intervals treat the replicate datasets within a stratum as independent Bernoulli trials, which is exact when the stratum is one configuration and approximate when it pools several: the pooled rate is then a mean over configurations with different true rates, and its interval is slightly narrower than sampling error alone would justify. The response curves are read for their trend rather than for the width of any one bin, and the Cochran–Armitage test used for that is computed on the same counts.

4.6 What the regime sweep adds

Three statements the calibration grid cannot make on its own come from grid B.

In the least confounded quintile of (5) — the runs where the channel contributes least — the outcome-chosen cutpoint is already at 0.267, more than five times the nominal level, while the cell-level test in the same quintile is what a calibrated test should be. The cutpoint’s error is therefore not a part of the confound; it is a separate cost that the confound adds to, which is the distinction Section 2.6 turns on.

The co-detection-matched family is nominal throughout the sweep, and the expression-matched family with it. That does not contradict the boundary found in grid A: what breaks co-detection matching is a mismatch between the query set’s co-detection and its matched draws’, and sweeping the ceiling moves both together, so the sweep varies the channel with that mismatch held fixed.

The refuted candidate fails differently here, and the way it fails is the argument against it. Binned by the tie-break share on this sweep the cell-level rate *rises*, at $z = +5.23$ ($P = 1.7 \times 10^{-7}$) — the opposite of the $z = -8.73$ the same test gives on grid A. Neither sign is a property of the share: on this sweep the share is almost a relabelling of the ceiling ratio itself (Spearman $\rho = +0.91$, against $\rho = +0.54$ on grid A), so its marginal trend here is the ceiling ratio’s trend wearing its name.

4.7 The two candidates in one model

Binning is a coarse way to compare two candidate quantities, and the direct comparison is to put both in the same model of whether the run rejected. The magnitude of the implied correlation and the tie-break share are each standardised, so the coefficients are per standard deviation and comparable, and fitted by iteratively reweighted least squares over the effect-free runs of grid A, of grid B, and of the two pooled. A quantity that earns its place has a positive coefficient of its own and keeps it when the other is present. The pooled fit spans 1,680 runs; the product’s coefficient is +4.165 ($z = +20.27$, $P = 2.1 \times 10^{-91}$) and the share’s is -0.018, with $P = 8.3 \times 10^{-1}$, indistinguishable from zero once the product is in the model. Asked on its own the share is -0.184 ($P = 4.2 \times 10^{-4}$). The share’s marginal

coefficient also changes sign between the sweeps, which is the same instability the binned rates show, seen without the binning.

4.8 Cohorts and preprocessing

The two cohorts are the GEO series GSE176078 [18] (55,003 cells $\times$ 24,646 genes, retrieved 2026-09-15) and GSE161529 [27] (33,061 $\times$ 33,538, retrieved 2026-09-16); the accession pages are linked in Data availability. Both were processed with Seurat [34] into annotated matrices before this study’s scoring step: GSE176078 carries the published cell-type labels, in 16 compartments; GSE161529 is partitioned into 18 Leiden groups at resolution 1.0, of which the 17 with at least 200 cells are scored — the 18th holds 129 cells and is skipped — and a compartment above 4,000 cells is subsampled to 4,000 with a fixed seed. The GSE176078 matrix enters as the source study supplies it, log-normalised — a per-cell monotone transform cannot reorder genes within a cell, so ranks are unchanged by it — while the GSE161529 counts are library-size scaled and $\log_{10}$ -transformed in the scoring pipeline. The axis a compartment is tested against is the first principal component over its 3,000 highly variable genes, the conventional count [35], and the matrices are handled through the scanpy/anndata ecosystem [36]. Gene sets come from MSigDB v2024.1.Hs [26]: six collections (h.all, c2.cp.kegg_legacy, c2.cp.biocarta, c2.cp.reactome, c2.cp.wikipathways, c7.immunesigdb), a set kept when 5–500 of its genes overlap the panel, and collections larger than a cap (reactome 250, wikipathways 200, immune-sigdb 400) subsampled to the cap with a fixed seed. That yields 1,374 sets for GSE176078 and 1,378 for GSE161529.

4.9 Real-data benchmark

The benchmark scores 1,374 gene sets from 6 MSigDB collections (h.all, c2.cp.kegg_legacy, c2.cp.biocarta, c2.cp.reactome, c2.cp.wikipathways, c7.immunesigdb) [26] against the 55,003 cells of GSE176078 [18], partitioned by the published annotation into 16 compartments and capped at 4,000 cells each so that no one compartment dominates the runtime. The panel is the 24,646 genes measured; the AUCell ceiling is 1,233, fixed from the whole matrix rather than recomputed per compartment, because the channel’s opening is a property of the compartment’s depth and a per-subset ceiling would rescale it away. Ranks are cached once per compartment and every gene set and null draw is then scored from the cache.

Each (compartment, gene set) pair contributes the score, the tie-break share of (4), the achieved ceiling-to-depth ratio, the correlation of the score with depth and of the compartment’s dominant axis with depth — the two factors of (5) — and the association a researcher would report, the score with that axis. The axis is the first principal component of the compartment’s expression over its highly variable genes, taken with the sign that loads positively on its strongest gene: the gradient an analyst would reach for without further justification, which is what makes it the right stand-in for the biological axis a gene set is scored against. A matched null is drawn for the first `n_null_sets` gene sets of each compartment, since the null’s cost rather than the scoring is what makes a full sweep over 1,374 sets and 16 compartments impractical.

The same pairs carry a cross-implementation check, because the arithmetic of the tie-break is not specific to AUCell. UCell [2] is scored twice per pair: once at the ceiling its documentation ships as the default for 10x data (1,500) and once at the ceiling its own guidance implies for that compartment, which is the median number of genes the cells detect. The two score vectors are correlated across the compartment’s cells, and the reported quantity is the fraction of pairs on which that correlation falls below 0.99 — the pairs where the default is not merely a different number but a different ordering of cells. The pair’s inclusion probability is then asked whether that disagreement is predictable from the depth vector alone, before any outcome is consulted.

4.10 Audit of the residual

The audit holds three configurations fixed — strong confounding with a very sparse set, strong confounding with co-expression, and an unconfounded control — and draws 24 replicate datasets from each. Every dataset is tested three separate times with the same procedure: once on the simulator’s own target set, and once on each of 192 *pseudo-targets* sampled from the null builder’s own generator. A pseudo-target is exchangeable with the null draws by construction, so a procedure that is anti-conservative for the simulator’s targets and nominal for the pseudo-targets is indicting the targets’ construction rather than the matching.

The audit runs on simulated datasets of 800 cells detecting a median of 430 of 12,000 genes, with target sets of 8 genes — a ceiling-to-depth ratio of 1.40, which is the ratio measured in the CD8⁺ T compartment of the real dataset. That choice is load-bearing. An earlier run of this audit used a panel of half the size and the same depth, which put the ratio below one and closed the channel: the two confounded configurations then behaved exactly like the control, and the audit reported, correctly and uselessly, that the confounder did nothing. The ratio is recorded with every audit row so that a configuration which never opens the channel cannot be mistaken for a confounded one again.

The quantity compared across the three kinds of set is the dispersion of the realised correlation across replicate datasets, against the dispersion of the matched draws within a dataset. A procedure whose null has the right spread has these equal, and their ratio is what the Results report.

4.11 Software and implementation

The framework is released twice, in the two languages the field uses, with the same behaviour rather than merely the same intent. The Python package `sparsegs` supplies the rank cache, the three null families, the diagnostics and the four-step checklist; the R package `sparseGenSetCal` supplies the same surface. The agreement is asserted rather than claimed: the R package is an independent implementation of the same specification, and the cross-language check (`cross_language_check.py`) runs the R implementation against the working copy of the Python one: 216 diagnostic reports spanning every branch of the verdict rule, 30 fixed null vectors at three tail conventions whose Monte-Carlo P values, percentiles and null dispersions agree to one part in 10⁹, the pre-flight calibrations and the convenience API, and the structured records, which come out identical as text; verdicts agree on every report, and the tie-break keys agree to the integer. Both packages build on the standard scientific stack — NumPy [37] and SciPy [38] in Python, the Matrix package in R. An interactive application walks the four steps on bundled datasets and on a user’s own matrix, and a tutorial notebook introduces the null construction from first principles.

The design decision that makes the framework affordable is the rank cache: ranks are computed once per matrix, clipped at the ceiling, and every gene set and every null draw is then scored from the cache in time linear in the number of cells. Without it, a matched null drawn for each of the 1,374 gene sets is not a computation anyone would run.

Four entry points carry the framework in both languages: `construct_expression_matched_null` draws a null and reports the composition it achieved rather than the one it was asked for, `compute_calibrated_score` scores a gene set against one, `diagnostic_checklist` runs the four steps in order and stops where they say to stop, and `validate_tier_framework` checks a claim against the technical, internal and external tiers. Each returns a `calibration_object`: one structurally identical record holding the inputs, the measured quantities, the verdicts and the seed, so that a calibration can be archived, compared between gene sets and re-run rather than only read. Every diagnostic is written as a record with named fields rather than a sentence, which makes a run countable — how many sets had to be drawn from a widened background is a question with an answer rather than an impression. For the reader who wants the decision without the machinery, two pure functions

restate the table and the fit: `select_null_method` returns the branch a set’s measured quantities put it on, with the thresholds it applied echoed in the result — generated from the same constants as Supplementary Table S5, so the function and the table cannot disagree — and `predict_fpr` returns the conventional test’s rejection probability under a true null for a given magnitude of the implied product, carrying the fit’s provenance and flagging any input beyond the range the fit saw.

The defaults are the ones the work supports. A set is flagged when its observed zero rate reaches `sparsity_threshold` (0.50); a null is accepted only within `null_matching_tolerance` (0.05) of the observed detection rate and expression; tests are read at `auc_significance_level` (0.05); the external tier asks for `replication_cohorts` (2) independent cohorts; cutpoint false-positive rates are permuted `cutpoint_permutations` (1000) times; and `seed` (42) makes a calibration object reproducible exactly.

Data availability

The two cohorts are public GEO series: GSE176078 [18] (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE176078>, retrieved 2026-09-15) and GSE161529 [27] (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE161529>, retrieved 2026-09-16). Gene sets are MSigDB v2024.1.Hs [26]. All simulated data are generated by the code in this repository from seeds recorded with each result: the five grids comprise 2,800 JSON-line records (1,080 in grid A, 600 in B, 480 in C, 400 in D, 240 in E), delivered with the submission as compressed archives, and `results_md5.txt` checksums every file under `results/` so that the exact analysed versions are verifiable. The analysis outputs quoted here — `numbers.tex` and every generated table — are regenerated from those records by `analyse_grid.py` and `make_numbers.py`; Supplementary Table S8 shows the most recent such check.

Code availability

The Python package `sparsegs`, the R package `sparseGenSetCal`, the interactive application, the tutorial notebook and the scripts that produce every number and figure in this paper are in the project repository, and both packages travel with the submission as source archives (`sparse_gene_set_calibration-0.1.0.tar.gz` and `sparseGenSetCal_0.1.0.tar.gz`). The simulation grids’ records travel as a compressed archive attached to the repository’s release record; `results_md5.txt`, kept in the repository, checksums every file of that archive. The claim that every table regenerates from the raw records was executed rather than assumed: on 2026-09-20 the generation chain was re-run over the archived records and reproduced `numbers.tex` and all six generated tables byte-for-byte, and Supplementary Table S8 recomputes the headline rates from the records independently of the generated macros, with exact agreement. The test suites passed in full on the same day — 127 pytest tests and 51 R test blocks, none skipped — together with the cross-language check described in Methods. The continuous-integration workflow is defined in the repository (`.github/workflows/ci.yml`) but has not yet run on a public host, so no CI status is claimed until the repository is public and the badge is green. The archived release carries a DOI, inserted here once minted: XXX-DOI-XXX.

Author contributions

M.T., X.C. and Q.D. designed the study and directed the analysis; all authors reviewed the code and the text and take responsibility for them.

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Competing interests

The authors declare no competing interests.

Key Points

- Rank-based gene-set scores of the AUCell type carry a sequencing-depth-driven tie-break channel whose size has a closed form.
- Each score decomposes exactly into a detected share, a tie-break share and a reproducible residual, so the depth contribution is quantifiable per cell.
- Expression- and co-detection-matched nulls absorb the channel across most of the practical range; the paper maps where the match stops holding.
- A four-step checklist with matched-null diagnostics and an explicit threshold tells practitioners when a sparse-panel score is safe to read.

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Effect	Matched family	Lower	Two-sided	Upper
<i>Grids A and B, no planted effect ($n = 1,680$ runs) — these are false-positive rates</i>				
	random-matched	0.035	0.199	0.210
	expression-matched	0.060	0.067	0.058
	codetection-matched	0.079	0.076	0.043
<i>Grid C, planted effect ($n = 240$ runs per effect) — these are power</i>				
0.25	random-matched	0.000	0.779	0.808
	expression-matched	0.000	0.871	0.908
	codetection-matched	0.000	0.863	0.900
0.5	random-matched	0.000	0.900	0.900
	expression-matched	0.000	0.988	0.992
	codetection-matched	0.000	0.988	0.996

Supplementary Table S1: The matched null in three conventions, computed from the same null draws in every case. The paper reports the two-sided column; the directional columns are given so that the choice can be checked rather than taken on trust. Reading them together is what separates the two ways a matched null can fail: a false-positive rate that depends on the tail is a defect of the test, while a power that the lower tail cannot reach is not a defect at all but the reason the convention has to be stated before the grid is run.

A Derivation of the closed forms

Order the ranks so that detected genes, which carry positive counts, occupy the top of the ranking in an order that is immaterial here, and the $G - D$ undetected genes tie for every remaining position. The tie-break assigns those tied genes an order drawn uniformly; the implementation's hash is one draw from that distribution, which is why a property of the distribution is a property of the score.

Inclusion. The top- m block holds all D detected genes and $m - D$ of the tied genes. A given undetected gene occupies one of the $G - D$ tied positions uniformly, so the probability that it lands inside the block is

$$\pi(D) = \frac{m - D}{G - D},$$

which is equation (1) of the main text.

Expected share. The AUC_{Cell} score is the mean, over ranks $i = 1, \dots, m$, of the fraction of the set recovered by rank i : $\text{AUC}_c = \frac{1}{mk} \sum_{i=1}^m n_i$, with n_i the number of set genes at rank $\leq i$ and $k = |S|$. Detected set genes contribute n_i up to their expression ranks — the Det_c of (3). An undetected set gene sits at rank $\leq i$ with probability 0 for $i \leq D$ and $(i - D)/(G - D)$ for $i > D$, by the same uniform-order argument. Its expected contribution to the score is therefore

$$\frac{1}{mk} \sum_{i=D+1}^m \frac{i - D}{G - D} = \frac{(m - D)(m - D + 1)}{2m(G - D)k} = \frac{\tau(D)}{k},$$

with $\tau(D)$ the closed form (2) of the main text. The undetected members of the set number $k - d_c(S)$, so their total expected share is $\frac{k - d_c(S)}{k} \tau(D)$ — the middle term of equation (3). The residual ε_c is the realised tie-break centred on this expectation: zero in the mean by construction, reproducible because the hash is deterministic, and its measured size is reported in the Results.

Cohort	Cells	Genes	Compartments	Ceiling	Median detected	Inclusion	Sets scored
GSE176078	55,003	24,646	16	1,233	1476	0.006	1374
GSE161529	33,061	33,538	17	1,677	1073	0.017	1378

Supplementary Table S2: The two real cohorts the benchmark runs on. *Ceiling* is the rank ceiling the 5% AUCell fraction implies for that panel; because it is a fraction of the panel, the same gene set lands on a different scale in each cohort, which is why scores are compared between cohorts only in direction and rank association. *Median detected* is the median over compartments of the median genes a cell detects there, *inclusion* the median over compartments of the tie-break inclusion probability, and *Sets scored* the sets retained after the usable-overlap filter.

B Supplementary tables

C Companion figure

D Reproducibility statement

Data. The two cohorts are public GEO series — GSE176078 (55,003 cells $\times$ 24,646 genes; retrieved 2026-09-15) and GSE161529 (33,061 $\times$ 33,538; retrieved 2026-09-16) — with the accession pages linked in Data availability; gene sets are MSigDB v2024.1.Hs. `results_md5.txt` checksums every file under `results/`, so the exact analysed versions are verifiable against an independent download.

Code. `sparsegs` 0.1.0 (Python) and `sparseGenSetCal` 0.1.0 (R) ship with full test suites. On 2026-09-20 the Python suite passed 127 tests and the R suite passed 51 test blocks with none skipped; the cross-language check of Methods ran clean the same day — 216 diagnostic reports with agreeing verdicts, 30 fixed null vectors at three tail conventions with P values agreeing to one part in 10^9 , 15 pre-flights, 7 structured records identical as text, and 15 convenience calls agreeing to one part in 10^{12} . Environments are pinned in `renv.lock` (R) and `environment.yml` (Python).

Verification. Every rate and table is generated from the raw JSON-line records by `analyse_grid.py` and `make_numbers.py`. On 2026-09-20 the generation chain was re-run over the archived records and reproduced `numbers.tex` and all six generated tables byte-for-byte; Supplementary Table S8 recomputes the headline rates from the records independently of the generated macros, with exact agreement.

Records. The five grids hold 2,800 records, each carrying its seed, its realised depth statistics and every test outcome. One record, abridged for width, reads:

```
{"grid": "A", "n_cells": 1200, "n_genes": 12000, "n_target": 8,
  "seed": 1034, "effect": 0.0, "depth_ratio": 2.094, "tie_break_share": 0.440,
  ..., "naive_reject": false, "expression_reject": false,
  "codetection_reject": false}
```

Runtime. The real-data benchmarks took 11,832 s on GSE176078 and 7,102 s on GSE161529 on the machine that produced them; the simulation grids run at roughly 0.3 configurations per second on one core, so a full regeneration of all five grids is an overnight job and parallelises across configurations.

Collection	Sets	Median k	Detection	Median $ \rho $	Pairs ≥ 0.3	Within-compartment ρ
<i>GSE176078</i>						
Hallmark	50	168	6.6%	0.349	57%	+0.64
Immunologic	400	185	6.4%	0.273	45%	+0.41
Reactome	250	26	7.0%	0.193	30%	+0.43
BioCarta	288	14	8.4%	0.175	22%	+0.30
KEGG	186	46	4.9%	0.173	27%	+0.26
WikiPathways	200	27	5.6%	0.162	22%	+0.28
<i>GSE161529</i>						
Hallmark	50	176	5.8%	0.496	77%	+0.72
Immunologic	400	193	5.1%	0.470	75%	+0.73
Reactome	250	23	5.2%	0.259	44%	+0.57
KEGG	186	52	4.0%	0.255	43%	+0.42
BioCarta	292	15	6.6%	0.248	39%	+0.48
WikiPathways	200	25	4.9%	0.245	42%	+0.35

Supplementary Table S3: Every collection the benchmark draws on, scored in all sixteen annotated compartments of GSE176078 and all seventeen of GSE161529, so a difference between rows within a block is a difference between sets rather than between the compartments that happen to carry them. *Collection* is the MSigDB collection the set names belong to, with the immunologic signatures grouped by elimination; *Sets* counts the sets retained after the usable-overlap filter, *Detection* is the median fraction of cells in which a set’s genes are seen (per cent), *Median $|\rho|$* the median absolute correlation of the score with the cell’s detected-gene count over that collection’s (compartment, set) pairs, and *Within-compartment ρ* the median over compartments of the Spearman correlation between a set’s median detection rate and its absolute score–depth correlation. That last column is the diagnostic’s within-compartment performance, and it has the same sign in every collection, every compartment and every cohort.

Continuous integration and archiving. The CI workflow is defined in the repository and will exercise the regeneration on a public host once the repository is released; until then no CI status is claimed. The archived release will carry a DOI (XXX-DOI-XXX pending minting). Questions about code, data or reproducibility go to the corresponding author, duanqiwen603@gmail.com.

Theme	Gene set	k	Detection	Median $ \rho $, 1st	Median $ \rho $, 2nd
Hypoxia	HALLMARK_HYPOXIA	189	5.4%	0.466	0.649
	GSE26023_PHD3_KO_VS_WT_	178	3.7%	0.318	0.515
Cell cycle	NEUTROPHIL_HYPOXIA_DN				
	BIOCARTA_P53HYPOXIA_PATHWAY	21	11.8%	0.134	0.420
	REACTOME_THE_ROLE_OF_	76	26.5%	0.504	—
	GTSE1_IN_G2_M_PROGRESSION_				
	AFTER_G2_CHECKPOINT				
	REACTOME_G2_M_CHECKPOINTS	127	10.3%	0.452	0.713
	KEGG_CELL_CYCLE	124	6.5%	0.302	0.558
	REACTOME_TP53_REGULATES_	20	11.8%	0.203	0.250
	TRANSCRIPTION_OF_				
	ADDITIONAL_CELL_CYCLE_				
	GENES_WHOSE_EXACT_ROLE_IN_				
	THE_P53_PATHWAY_REMAIN_				
	UNCERTAIN				
	HALLMARK_G2M_CHECKPOINT	190	7.3%	0.200	0.569
	HALLMARK_E2F_TARGETS	195	6.7%	0.159	0.491
	REACTOME_TRANSCRIPTIONAL_	34	8.3%	0.112	—
Immune checkpoint	REGULATION_BY_E2F6				
	REACTOME_PTK6_REGULATES_	6	8.9%	0.096	0.218
	CELL_CYCLE				
	BIOCARTA_SKP2E2F_PATHWAY	10	4.9%	0.085	0.251
	WP_MIR124_PREDICTED_	6	8.1%	0.064	—
	INTERACTIONS_WITH_CELL_				
Interferon	CYCLE_AND_DIFFERENTIATION				
	REACTOME_PD_1_SIGNALING	26	8.4%	0.262	0.205
Interferon	BIOCARTA_CTLA4_PATHWAY	20	5.6%	0.217	0.178
	HALLMARK_INTERFERON_	196	10.8%	0.490	0.526
	GAMMA_RESPONSE				
	WP_TYPE_II_INTERFERON_	34	14.7%	0.442	—
	SIGNALING				
	HALLMARK_INTERFERON_ALPHA_	95	12.4%	0.355	0.425
	RESPONSE				
	WP_TYPE_III_INTERFERON_	10	8.3%	0.184	0.199
	SIGNALING				

Supplementary Table S4: The named sets in the four themes a reader is most likely to test the framework on, selected from the collections by name rather than chosen: every set whose name carries a hypoxia, cell-cycle, immune- checkpoint or interferon term is here. *Median $|\rho|$* is the median absolute correlation of the score with the cell’s detected-gene count over the compartments of each cohort: the first is GSE176078, the second GSE161529, whose cells detect fewer genes against a higher ceiling (Table S2), so the same arithmetic predicts more damage there. A dash is a set that one cohort did not retain.

Measurement	Reading	What it means
Implied product $\rho_s \rho_a$ from the depth vector alone (pre-flight)	inside the 95% chance band	the channel is closed; the conventional test stands, and the diagnostic's output is the certificate that nothing needs doing
	outside the band	the channel is open; a conventional p -value is not evidence, whatever it reads
tie_break_share (step 1)	< 0.50	proceed to the null
	≥ 0.50	most of the score is tie-break; a matched null is required
	≥ 0.75	NOT_IDENTIFIABLE; report the refusal rather than a p -value
Drawn null's composition (step 2)	overlap 0, detection gap ≤ 0.05	the draw is a null; its p -value is readable
	either fails	the draw is not a null; no p -value it produces is evidence
Co-detection gap of the best draw, $ \rho(\text{axis}, \text{depth}) > 0.2$	≤ 0.005	matched families nominal (0.062)
	> 0.005	the co-detection family rejects at 0.187 on the same draws; prefer the expression-matched family and report the achieved gap
	> 0.020	no family is safe; the honest output is a refusal (0.228)
Cutpoint (step 3)	pre-specified chosen from the outcome	the nominal rate is readable measure the rate by permutation (1,000 draws) rather than assuming it
Independent cohort (step 4)	sign and rank association	direction only, never values; replication_cohorts = 2

Supplementary Table S5: The framework's decision points, as the software applies them. Every threshold is imported from the implementation rather than restated, so the table and the code cannot disagree; the three quoted rates are the grid's measured values from Table 4. A reading the framework cannot take — a cache without depth, falling back on the fixed detection floor — weakens the verdict and is flagged as such in the output.

Tool	Ranking	Ceiling	Where the channel sits	Measured here
AUCell [1]	every gene of the panel, within the cell	released default [0.05 G]	empty top ranks are filled by tie-break: the subject of this paper	yes — the grids and the benchmark
UCell [2]	every gene, within the cell	ships 1,500 for 10x data; its guidance recommends the median detected	the same fill, at whatever ceiling the call passes	yes — default against recommendation, per pair
singscore [4]	every gene, within the cell	no truncation	undetected genes tie at the bottom of the ranking; the fill does not arise, and the arbitrary order inside the tied block is a different exposure	no
GSVA [3]	every gene, kernel-weighted	no truncation	bottom-block ties as in singscore, smoothed by the kernel	no
module score [5]	none — a mean of expression values	—	no ranking; depth enters through normalisation	no

Supplementary Table S6: Where the tie-break channel sits in the ranking tools the introduction names. The *Ceiling* column quotes each tool’s own default where one exists: AUCell’s from the released Bioconductor manual, checked against the current release; UCell’s from its documentation, as in the Results. Rows marked *no* in the last column are statements about the published algorithms, read rather than measured; the table keeps that distinction visible instead of blurring it.

Grid	Runs	Configurations	Cells	G	m	Factors varied (levels)	Held fixed	Seeds
A	1,080	18 (60 each)	1,200	12,000	600	programme–depth loading: 0, 0.3, 0.6; per-gene detection: 1%, 5%, 15%; planted co-expression: 0, 0.6	planted effect 0; target-set size 8; donors 1	1,000–2,003
B	600	2 (300 each)	1,200	12,000	600	programme–depth loading: 0, 0.6	per-gene detection 5%; planted co-expression 0; planted effect 0; target-set size 8; donors 1	1,000–1,833
C	480	12 (40 each)	1,200	12,000	600	programme–depth loading: 0, 0.6; per-gene detection: 1%, 5%, 15%; planted effect: 0.25, 0.5	planted co-expression 0; target-set size 8; donors 1	1,000–1,663
D	400	8 (50 each)	1,200	12,000	600	per-gene detection: 1%, 5%; target-set size: 4, 8, 20, 50	programme–depth loading 0.6; planted co-expression 0; planted effect 0; donors 1	1,000–1,833
E	240	6 (40 each)	1,200	12,000	600	programme–depth loading: 0, 0.6; donors: 1, 4, 16	per-gene detection 5%; planted co-expression 0; planted effect 0; target-set size 8	1,000–1,663

Supplementary Table S7: The five simulation grids as their own records state them: 2,800 runs in all. A factor is varied when the grid’s records disagree on it, with the observed levels; the held-fixed column carries the factors every run of that grid shares. Cells, panel size G , ceiling m and the seed range are counted from the records, so the table is a complete specification of what was run rather than a summary of the columns that moved. The seed base and the replication rule are stated in the Methods.

Supplementary Table S8: The published rates, recomputed from the raw records. On 2026-09-20, the six headline false-positive rates of grid A were re-derived from the archived records in `grid_A_calibration.jsonl` with the analysis pipeline’s own rate function, and compared with the values the manuscript quotes. Every recomputed rate equals the published macro; the largest absolute difference is 0.000. The regeneration that produced `numbers.tex` and every generated table was re-run the same day and reproduced them byte-for-byte.

Analysis	Published	Recomputed	(k of n)
Cell-level correlation	0.400	0.400	(432 of 1,080)
Label permutation	0.397	0.397	(429 of 1,080)
Median split	0.337	0.337	(364 of 1,080)
Cutpoint chosen from outcome	0.556	0.556	(601 of 1,080)
Expression-matched null	0.068	0.068	(73 of 1,080)
Co-detection-matched null	0.094	0.094	(102 of 1,080)

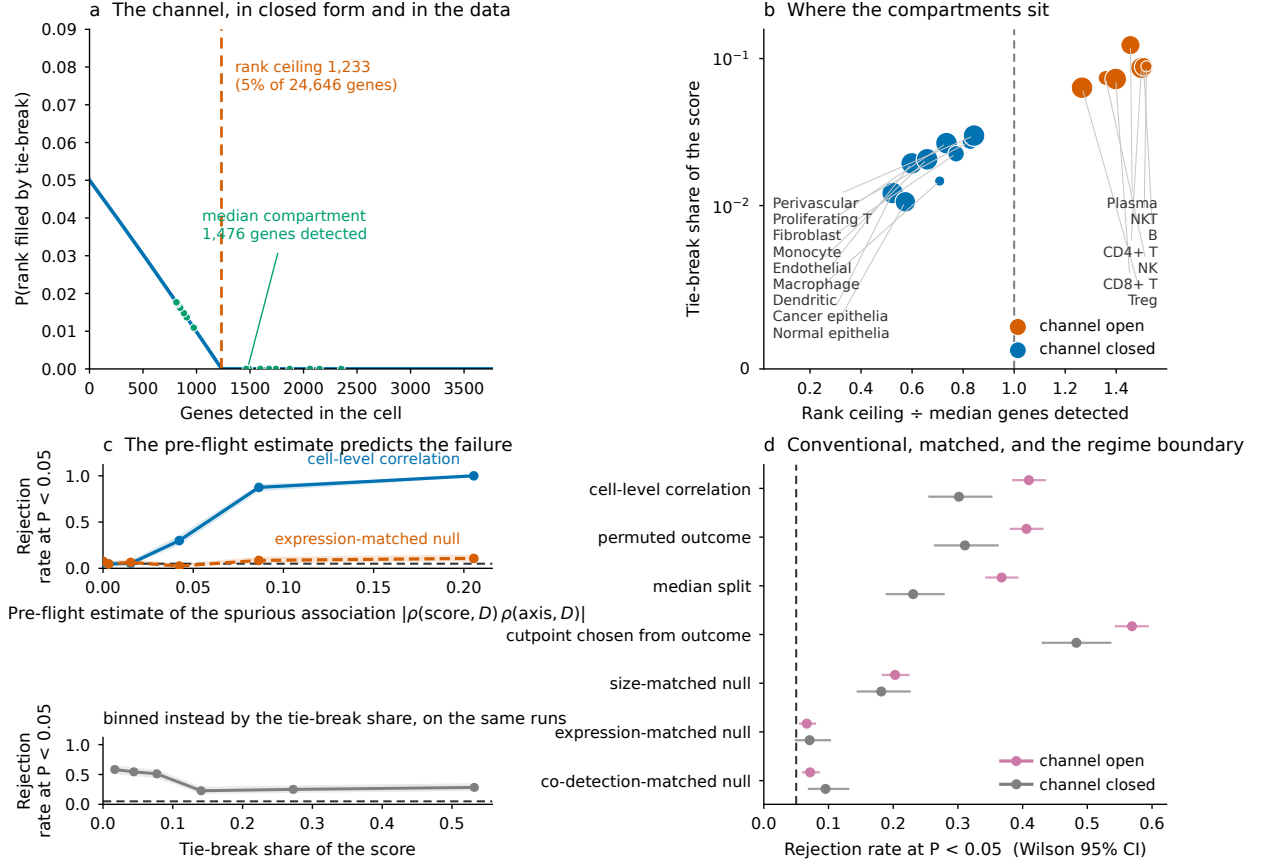


Figure 1: The argument in four panels. (a) Equation (1) for the panel and ceiling of this study, with each compartment of GSE176078 marked at its own median depth. (b) The regime against what it does. The ceiling-to-depth ratio is plotted against the share of the score that came from ranks the tie-break settled rather than from counts the cell observed, per compartment: the x axis is a property of the compartment’s depth, the y axis is what the score did there, and compartments to the right of the vertical line have the channel open. (c) Two rows on a shared vertical scale, both over the effect-free runs of the two calibration sweeps: above, the rejection rate in quantile bins of (5), for the cell-level test and the expression-matched null; below, the same runs binned instead by the tie-break share the paper first proposed, which falls where the argument requires it to rise. The dashed line in both rows is the nominal $\alpha = 0.05$. (d) All seven analyses, split by whether there is a channel to be confounded by.